

SPRING INTERNSHIP PROJECT REPORT

Time-Series Analysis of COVID-19 Data

Raunak Sen,

Programming and Data Science

Indian Institute of Technology Madras

Period of Internship: 21ST January 2026 – 17TH February 2026

Report submitted to: IDEAS – Institute of Data
Engineering, Analytics and Science Foundation, ISI
Kolkata

1. Abstract

This project focuses on performing a comprehensive time-series analysis of global COVID-19 data to understand pandemic trends across countries and WHO regions. The study uses real-world datasets to explore patterns in confirmed cases and deaths over time. Data preprocessing techniques were applied to clean and transform raw data into a usable analytical format. Time-based aggregation such as daily, monthly, and quarterly analysis was performed to observe trend variations. Exploratory Data Analysis (EDA) was conducted using statistical summaries. The project identifies highly affected countries and peak infection periods during the pandemic. Regional comparisons were also carried out to understand global spread patterns. Python and Pandas were used as the primary analytical tools. The findings help demonstrate how time-series analysis can support public health insights and decision-making. This project highlights the importance of data analytics in understanding real-world global events.

2.Introduction

The COVID-19 pandemic generated a vast amount of global data related to confirmed cases, deaths, and regional spread patterns. Analyzing such real-world datasets helps in understanding how infectious diseases evolve over time and across geographical regions. Time-series analysis plays an important role in identifying trends, peak infection periods, and comparative regional impacts, making it a valuable tool in data science and public health analytics.

This project, titled *Time-Series Analysis of COVID-19 Data*, was carried out during the Spring Internship conducted by IDEAS – Institute of Data Engineering, Analytics and Science Foundation, ISI Kolkata. The project focuses on applying data analytics techniques using Python to preprocess, analyze, and interpret global COVID-19 datasets. The workflow involved data cleaning, transformation of date variables, temporal filtering, aggregation of cases across different time intervals, and regional comparison using statistical methods.

The purpose of this project is to demonstrate how time-series analytical techniques can extract meaningful insights from real-world datasets. By performing exploratory analysis and structured aggregation, the study aims to understand pandemic progression patterns and highlight the importance of data-driven decision making.

Training Topics Covered During the First Two Weeks of Internship

During the initial training phase, several foundational and advanced topics were covered to prepare interns for data analysis projects:

Python Programming Fundamentals

- Data types, variables, and control structures
- Lists and loops
- Data structures in Python
- Functions, classes, and Object-Oriented Programming (OOP)

Data Analysis Tools

- NumPy for numerical computation
- Pandas for data manipulation and analysis

Machine Learning Concepts

- Overview of Machine Learning
- Regression techniques
- Classification techniques

Advanced & Professional Skills

- Fundamentals of Large Language Models (LLMs)
- Communication and presentation skills

3.Project Objective

The main objectives of this project are:

- To analyze global COVID-19 case trends over time using time-series analysis.
- To identify the most affected countries based on cumulative confirmed cases.
- To perform temporal aggregation such as daily, monthly, and quarterly analysis.
- To compare pandemic impact across WHO regions.
- To generate structured analytical tables and insights for reporting purposes.

4.Methodology

The project followed a structured data analysis workflow starting from data collection and preprocessing to analysis and interpretation of results.

Data Collection

A real-world global COVID-19 dataset, provided as part of the internship program, was used for analysis. The dataset contained information related to confirmed cases, deaths, countries,

WHO regions, and reporting dates. This structured dataset enabled detailed temporal and regional analysis of pandemic trends.

Data Preprocessing

Several preprocessing steps were performed to prepare the dataset for analysis. The dataset was imported using Python in the Google Colab environment. The data was examined for missing or inconsistent values to ensure accuracy and reliability. Date columns were converted into datetime format to support time-series operations. The dataset was then filtered according to the selected pandemic timeline, and column formats were standardized to maintain consistency during analysis.

Data Analysis Process

The analytical workflow began with Exploratory Data Analysis (EDA) to understand the structure and characteristics of the dataset. Highly affected countries were identified based on cumulative confirmed case counts. Daily global case trends were calculated to observe variations over time. Further analysis involved aggregating confirmed cases and deaths on monthly and quarterly bases to identify broader temporal patterns.

Regional comparisons were performed using WHO region grouping to understand geographical variations in pandemic impact. The peak global infection day was identified by analyzing daily case totals. Additionally, pivot tables were created to enable comparative regional analysis and generate structured summaries for interpretation.

Tools and Technologies Used

The project was implemented using Python as the primary programming language. The Pandas library was used extensively for data manipulation and aggregation. Google Colab served as the execution environment, and GitHub was used for maintaining version control and project documentation.

Model Development

No machine learning predictive model was developed in this project. The analysis focused primarily on descriptive analytics and time-series aggregation techniques to understand historical pandemic trends.

Code Repository

The GitHub repository containing the project code and documentation has been provided in the Appendix section for reference.

5. Data Analysis and Results

The analysis produced several important insights regarding global COVID-19 trends.

Descriptive Analysis

- Daily global case counts revealed clear waves of infection.

```
Date with highest global new cases: 2022-01-30 00:00:00
```

```
Number of cases on that day: 8401963.0
```

```
daily_global_cases = (df_covid_trimmed.groupby('Date_reported')['New_cases'].sum())  
daily_global_cases.head()
```

New_cases	
Date_reported	
2020-03-01	2003.0
2020-03-02	1940.0
2020-03-03	1796.0
2020-03-04	2215.0
2020-03-05	2165.0

dtype: float64

Fig. 1

- Certain countries consistently reported higher cumulative cases.

New_cases	
WHO_region	
EUR	222061547.0
AMR	158832865.0
WPR	67373243.0
SEAR	52144146.0
EMR	21806594.0
AFR	9042978.0
OTHER	65.0

Fig. 2

- According to Fig. 4. WHO regional aggregation showed uneven pandemic distribution.

Time-Series Findings

- Monthly and quarterly aggregation helped identify growth phases.(Fig.4 & 5)
- Peak infection periods were detected using global daily totals.(Fig. 5)
- Pivot tables enabled comparison of regional trends across time.

Visualizations

The project generated:

- Regional comparison tables in descending order

```
cases_by_region = (df_covid_trimmed.groupby('WHO_region')['New_cases'].sum().sort_values(ascending=False))
cases_by_region
```

New_cases	
WHO_region	
EUR	222061547.0
AMR	158832865.0
WPR	67373243.0
SEAR	52144146.0
EMR	21806594.0
AFR	9042978.0
OTHER	65.0

dtype: float64

Fig. 3

- Aggregated summary tables by Quarter

```
df_covid_trimmed['Quarter'] = df_covid_trimmed['Date_reported'].dt.to_period('Q')
quarterly_data = (df_covid_trimmed.groupby('Quarter').agg({'New_cases': 'sum', 'New_deaths': 'sum'}))
quarterly_data.head()
```

	New_cases	New_deaths
Quarter		
2020Q1	697567.0	40666.0
2020Q2	9377665.0	511574.0
2020Q3	23887893.0	546584.0
2020Q4	48287242.0	834992.0
2021Q1	45249877.0	1026874.0

Fig. 4

- Monthly pivot tables

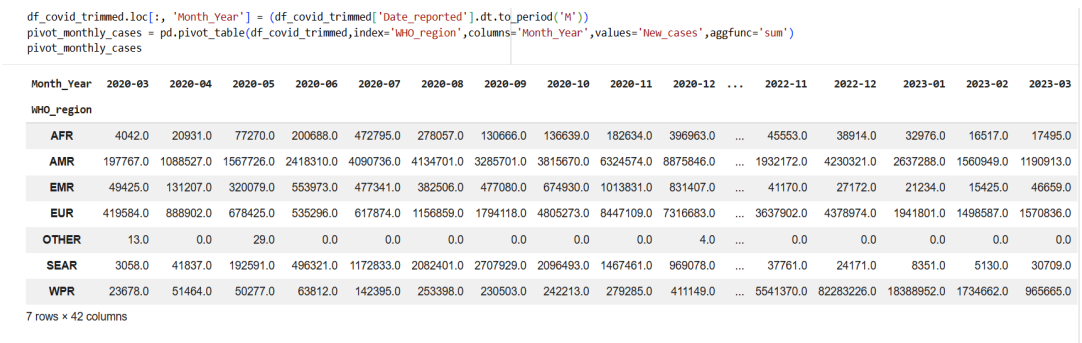


Fig. 5

Key Results

- Ranking of highly affected countries.



Fig. 6

- Regional comparison of pandemic spread patterns.

6. Conclusion

This project successfully demonstrated how time-series analysis can be applied to real-world pandemic data to extract meaningful insights. Through data cleaning, aggregation, and exploratory analysis, important global trends and regional variations were identified. The findings show that structured data analysis helps in understanding disease progression and supports informed decision-making. The project also strengthened practical skills in data preprocessing, aggregation, and analytical reporting using Python.

7. APPENDICES

Appendix A – References

- Python Pandas Documentation

- Online research articles and analytics resources

Appendix B – GitHub Repository

https://github.com/RaunakSen-EE/06-Time-Series_Analysis_of_COVID-19_Data_Spring_2026.ipynb

Appendix C – Google Colab Notebook

https://colab.research.google.com/drive/1x6ucXaqY09eegPtGsigBjyy5YOX2_LrV?usp=sharing#scrollTo=q9-code

Appendix D – Additional Documents

- Project Report
- Dataset files
- Presentation slides